

JOHNATHAN AHDOUT

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EDUCATION

Princeton University

B.A. Mathematics; Minors: Computer Science & Optimisation and Quantitative Decision Science

Princeton, NJ
Present – May 2029

- GPA: 3.79/4.00
- Relevant Coursework: *Honours Linear Algebra, Real Analysis, Probability Theory, Data Structures & Algorithms, Financial Investments & Derivatives, Abstract Algebra, Combinatorics, Fundamentals of Statistics*

John L. Miller Great Neck North High School

Salutatorian, GPA: 97.8/100, SAT: 1590 (800 Math, 790 Reading & Writing)

Great Neck, NY
Sep 2021 – Jun 2025

RESEARCH & EXPERIENCE

Deep Ocean Partners

Quantitative Research Intern — Private Credit & Customer-Base Modelling

New York, NY (Remote)
Sep 2026 – Present

- Model customer purchasing & churn behaviour as a Poisson process (Pareto/NBD, BG/NBD) in Python to forecast borrower revenue for private-credit underwriting.
- Built a hierarchical Bayesian extension (PyMC) adding seasonality, cohort-quality & discount effects; validated on held-out data, then presented retention curves and revenue forecasts to the firm's Head of Data Science.

Dickson Lab @ Michigan State University, ACRES REU

NSF-Funded Research Intern — Generative AI for Molecular Design

East Lansing, MI
May 2026 – Present

- Extended the LEVI molecular generation framework (PyTorch) with EGNN & diffusion architectures for protein-conditioned generation & inpainting.
- Developed spectral graph-theoretic forces improving molecular connectivity **30–50%** while preserving quality; extending them with network flows & bipartite matching.
- MSU Mid-SURE ([poster](#)); **1st Place**, ACRES Symposium. **Second author** on *Learning Composable Time-Unconditional Molecular Energy Landscapes for Zero-Shot Environmental Transfer* (ICLR, in prep.); **first author** on a proof-based diffusion manuscript.

Independent Research, Princeton University

Conditional Mutual Information for Point-in-Time Macro-Quantamental Screening

Princeton, NJ
Aug 2026 – Present

- [Proposed](#) screening J.P. Morgan JPMaQS macro indicators for predictive content beyond linear measures via conditional mutual information; derived the bias–variance rate of a kNN CMI estimator & a cross-country panel-pooling procedure relaxing its dimensionality limit. With Macrosynergy & Princeton ORFE researchers.

Princeton University, Department of Mathematics

Undergraduate Course Assistant — MAT 215 (Single-Variable Analysis with Proofs)

Princeton, NJ
Sep 2026 – Present

- Run weekly office hours and grade problem sets for Princeton's proof-based real-analysis course.

EXTRACURRICULAR INVOLVEMENT & PROJECTS

Princeton Puzzles Club

Officer & Princeton Puzzlehunt Puzzle Writer/Tester

Sep 2025 – Present

- Design & test competition-grade logic puzzles, constructing multi-step solution paths and verifying uniqueness; wrote and tested three puzzles for Princeton Puzzlehunt 2026 (500+ participants).

Princeton Quantitative Traders

Member & Data Science Workshop Educator

Sep 2025 – Present

- Teach options pricing, statistical modelling & quant-finance ML in Python; build and evaluate trading heuristics (Kelly optimisation, spread/latency modelling) in mock trading competitions.

Princeton Game Theory Club

President (previously Events Director, Strategy Team, Newsletter Editor)

Sep 2025 – Present

- Organise meetings and tournaments on iterated games & adversarial reasoning; wrote the club rulebook and $\text{\LaTeX}$ strategy guide, and won the inaugural GTC Fish competition (Fall 2025).

Options Pricing & Volatility-Trading Research Toolkit (Python) | [Live Dashboard](#)

Jun-Aug 2026

Independent Project

- Built three options-pricing engines (Black–Scholes, Monte Carlo, CRR binomial tree), with Newton–Raphson/Brent/Jaekel IV solvers benchmarked across 250 cases.
- Derived a gamma/theta P&L identity from Itô's lemma; implemented a Gaussian HMM (Baum–Welch EM, Viterbi) for volatility-regime detection, backed by a 97-test suite and a live SPY implied-vol surface; shipped the whole toolkit as a five-tab [Streamlit dashboard](#).

GridSmart: Real-Time Grid Optimisation & Smart Home Scheduling

Apr 2026

Princeton University Claude Builder Club Hackathon

- Built an AI platform (ERCOT pricing, SmartThings, Alexa APIs) auto-scheduling home energy use by cost & grid demand; [presented](#) to Princeton ORFE & CS faculty.

SKILLS, AWARDS & CERTIFICATIONS/DISTINCTIONS

Languages & Tools: Python, Java, R, PyTorch, PyMC, scikit-learn, NumPy, Pandas, $\text{\LaTeX}$, Excel | [J.P. Morgan Quantitative Research Job Simulation](#) (Forage)

Honours:

- National Merit \$2500 Scholar | U.S. Presidential Scholar Semifinalist | Coca-Cola Scholar Semifinalist (all 2025)
- [Regeneron STS 2025 Top 300 Scholar](#) — QSPR property prediction, in the Princeton Undergraduate Research Journal
- Co-author, *PLOS Biology*: [SARS-CoV-2 envelope-protein inhibitory peptides](#) | Hagan Scholarship Finalist (2025)
- Quiz Bowl NAQT Nationals (2024, 2025), Princeton A-Team Top 10 | 2× IAC Science Bee Nationals, 15th & 4th